

A Quick Look At Free Platforms And Libraries For QUANTUM MACHINE LEARNING

Quantum computing, due to its ability to calculate at an immense speed, has the potential to solve many problems that classical computers find difficult to address. Quantum machine learning, or QML, is a new field that explores the intersection between quantum computing and machine learning. Several libraries and platforms facilitate the development of QML algorithms and applications. A few popular ones are discussed in this article



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Quantum computing uses quantum mechanics to perform calculations. While classical computers use bits, which can represent either 0 or 1, quantum computers use qubits, which can exist in multiple states simultaneously. This allows quantum computers to perform certain types of calculations much faster than classical computers, especially those related to optimisation, machine learning, and cryptography.

However, building quantum computers is a significant technical challenge, as qubits are highly sensitive to environmental noise and require sophisticated error-correction techniques. Despite these challenges, there is significant interest in the potential applications of quantum computing in areas such as drug discovery, materials science, and artificial intelligence.

Quantum computing has the potential to revolutionise many fields by solving problems that are currently intractable using classical computing. There is a huge scope for quantum computing in assorted domains, and it has the potential to impact nearly every field of science and technology. However, the technology is still in its early stages of development, and there are significant challenges to be addressed in terms of hardware limitations, error correction, and algorithm development, as well as scalability and reliability. Nonetheless, with continued research and development, quantum computing could lead to significant breakthroughs and advancements in many areas of science and technology.

In 2020, Google claimed to have achieved quantum supremacy with its Sycamore quantum computer, which took